Aido Rover-Class Platform Power Budget & Mission Time Estimation

1. Battery Configuration (4S2P 18650 Pack)

Parameter	Value	Unit	Notes / Formulas
Battery Chemistry	Lithium-Ion (18650)	-	Standard high-density cells
Series Cells (S)	4	-	Determines pack voltage
Parallel Cells (P)	2	-	Determines pack capacity
Nominal Cell Voltage	3.6	V	Standard nominal cell voltage
Nominal Cell Capacity	3.5	Ah	Assumed 3500 mAh per cell
Usable Capacity Factor	80%	-	Requested operational depth of discharge
Pack Nominal Voltage	**14.4**	**V**	**= 4 * 3.6 V**
Pack Total Capacity	**7.0**	**Ah**	**= 2 * 3.5 Ah**
Total Nominal Energy	**100.8**	**Wh**	**= 14.4 V * 7.0 Ah**
Total Usable Energy	**80.64**	**Wh**	**= 100.8 Wh * 80%**

2. Component Power Consumption Breakdown

Component	Nominal Power (W)	Peak Power (W)	Duty Cycle (%)	Average Power (W)
Compute (Jetson Orin Nano)	10	15	100%	10.0

Component	Nominal Power (W)	Peak Power (W)	Duty Cycle (%)	Average Power (W)
Drive Motors (Base & Actuators)	25	60	60%	15.0
Sensor Stack (LiDAR, Cameras, IMU)	12	18	100%	12.0
Communications (Wi-Fi + LTE)	5	10	70%	3.5
Payload Allowance	15	30	50%	7.5

3. Power Budget Summary

Metric	Value	Unit	Calculation / Description
Continuous Power Draw	22.0	W	Compute (10W) + Sensors (12W)
Average Dynamic Power Draw	26.0	W	Motors (15W) + Comms (3.5W) + Payload (7.5W)
Total Average System Power	**48.0**	**W**	**Continuous (22W) + Average Dynamic (26W)**
Total Peak System Power	**133.0**	**W**	**Sum of all peak power limits**

4. Operational Mission Time Estimation

Parameter	Value	Unit	Formula
Total Usable Energy Available	80.64	Wh	From Battery Configuration
Total Average System Power Draw	48.00	W	From Power Budget Summary
Estimated Mission	**1.68**	**Hours**	**= 80.64 Wh / 48.0 W**

Parameter	**Value**	**Unit**	**Formula**
Time (Hours)			
Estimated Mission Time (Minutes)	**100.8**	**Minutes**	**= 1.68 * 60**
Final Formatted Runtime	**1 Hour 40 Minutes**	**-**	**Operational time at 80% capacity**